

CHATBOTS

Minnah Jafar

INTRODUCTION

- Developed two AI-based applications:
 - AI Chatbot
 - RAG PDF Chatbot
- Designed to improve communication and document search efficiency.
- Built with a business-focused approach for practical real-world use cases.

AI CHATBOT

PROJECT OVERVIEW

- Developed an AI chatbot capable of generating human-like responses.
- Supports multi-turn conversations and context-based interaction.
- Built using Python, Groq API, and Streamlit.

PROJECT

1

BUSINESS PURPOSE

- Reduces repetitive manual communication
- Provides faster customer interaction
- Improves response efficiency
- Helps automate communication systems

KEY FEATURES

- AI-generated responses
- Multi-turn conversation support
- Chat history management
- Adjustable response settings
- Fast response generation using Groq AP

TECHNOLOGIES USED

- Python
- Groq API
- Streamlit



MY CONTRIBUTIONS

- Developed chatbot logic using Python
- Integrated Groq API
- Managed conversation flow and history
- Tested chatbot using different AI settings
- Improved response quality and interaction handling

Challenges Faced

- Understanding API integration
- Managing chatbot memory
- Tuning AI response quality

Solutions Implemented

- Experimented with temperature and max_tokens
- Improved prompt handling
- Structured conversation storage

CHALLENGES

AND

SOLUTIONS

RAG CHATBOT

PROJECT OVERVIEW

- Developed a RAG-based chatbot capable of answering questions from PDF documents.
- Uses Retrieval-Augmented Generation for intelligent document search.
- Built using LangChain, ChromaDB, Streamlit, and Groq API.

PROJECT

2

BUSINESS PURPOSE

- Helps users retrieve information quickly from documents
- Reduces time spent manually searching PDFs
- Improves productivity and accessibility
- Enables intelligent document interaction

KEY FEATURES

- PDF upload support
- Text extraction from PDFs
- Text chunking
- Embedding generation
- ChromaDB vector storage
- Intelligent retrieval system
- AI-generated answers



MY CONTRIBUTIONS

- Implemented PDF processing workflow
- Built retrieval pipeline using LangChain
- Integrated ChromaDB vector database
- Connected LLM for answer generation
- Designed Streamlit chatbot interface

Challenges Faced

- ChromaDB permission issues
- Streamlit rerun handling
- Managing uploaded PDFs

Solutions Implemented

- Improved project structure
- Optimized retrieval workflow
- Better session handling

CHALLENGES

AND

SOLUTIONS